## Vinayak Chatterjee on linking cities, boosting economy

# 'High-speed rail is the big move India should be looking at now'

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MORE Indians are flying than ever before. But in the process is air travel getting too clunky and burdensome? Perhaps it is time to balance the emphasis on flying with fast and efficient train services over short inter-regional distances.

An opportunity beckons in linking up cities big and small and in the process attracting investment in industries that might not otherwise head in these directions.

Some progress has already been made with the Vande Bharat trains. A bigger and calibrated push is in order. Dedicated railway corridors and superior technology could dramatically ease travel, speed up journeys, spur urbanization and join the dots that bring sizeable economic advantages.

Vinayak Chatterjee has for some time now been advocating high-speed rail services as the next big idea that Indian policy mandarins must embrace. High-speed rail would be a substantial improvement over current services and a notch or two below the bullet train. Preparations being made for the bullet train could easily be extended to ushering in high-speed rail.

Chatterjee has been an innovator in the infrastructure space for the longest time. For someone with large sums on his mind, he is also surprisingly inclusive and socially driven. Having built Feedback Ventures with friends and sold it off recently, he and his wife, Rumjhum, have co-founded the Infravision Foundation. Below is an edited version of a long and interesting conversation:

**Q: You have put forward the view that high-speed rail is going to be a game changer for the Indian economy. Tell us three good reasons why.**

I think the most compelling economic reason — if you look back on countries in Europe and other countries that started high-speed rail perhaps two or three decades ago — is that their per capita GDP was close to where India's is now. Japan, of course, started much earlier.

Therefore, there comes a time in a society and an economy when you have a larger section of people demanding better services. Not just, you know, electricity, tap water, cooking gas and all of that. The desire of society is to move up in terms of cheaper air connectivity, better utilities, public transport…

There is a national desire that our creaking railway system should address this need for faster and more comfortable travel. India's economic indices, its development aspirations, the aspirations of its people, its per capita income, all indicate we are now ready to start high-speed train services.

Civil Society picture/Umesh Anand


*Vinayak Chatterjee: 'We are losing customers willing to pay more for high-speed rail'*

The second reason is that the railways subsidize passenger traffic to the extent of ₹64,000 crore per annum. It is interesting, when you segment it, that a fair chunk is for AC two-tier, AC three-tier, AC first class, and executive chair car in the Shatabdi trains. That segment of the population is rapidly moving to air travel or personal vehicles or luxury Volvo inter-city coaches. So you are actually losing a large chunk of customers who are willing to pay three or four times more for a high-speed train.

And if you continue to run creaky trains then, in marketing and business terms, you are allowing a large segment of your well-heeled customer base to migrate from your service. Does it make any sense?

Third, we are arguing for high-speed rail on dedicated lines, bypassing existing lines. A lot of technical studies in the railways have shown that there is very high choking of trunk routes and junctions. Probably the most congested junction is Mughalsarai, now called Deen Dayal Upadhyaya Junction, where a cluster of movement on limited capacity takes place. There is coal movement, iron ore movement, passenger traffic, container traffic, the eastern corridor — all crunched together on limited capacity.

So if you take a whole lot of high-speed rail and passengers on a separate track, you will free up capacity for freight trains as well as for existing trains, right?

Already, a chunk of the population has moved to airlines and low-cost air travel. It is inevitable that people will want to move to high-speed rail. That itself opens up an opportunity for next-generation manufacturing. Obviously, there is a bunch of choices: semiconductors, chips, nuclear, electronic assembly of mobile phones, etc., in the manufacturing sector.

But if you look at the infrastructure sector, there have been two major national programmes. One, the National Highway Development

Programme. Two, the switch to renewables.

**Q: What has been their impact?**
Both have transformed the Indian economy. Today, half of India's generating capacity is in renewables. Out of 400 gigawatts of our installed capacity, 200 gigawatts is renewable capacity, which is solar, wind, hydro.

This combination of highways and renewable energy has, in the past two decades, probably led to investments of around ₹100 lakh crore, which is no laughing matter. Infra investments yield a multiplier of three and have galvanized the economy, according to my back of the envelope calculations, to the extent of maybe ₹300 lakh crore.

What is the next big idea that India has? To my mind, it is high-speed trains. Seven corridors of around 700 kilometres each have been identified by the rail ministry. We, in Infravision Foundation, have prioritized four in our report. The point is that seven such corridors across 15 years have the capacity to again galvanize about ₹20 lakh crore of investments with a three times multiplier effect and create a huge number of jobs both in the laying of tracks and associated systems as well as in the indigenization of manufacture of all equipment from coaches to signalling to stations to safety. It's a massive programme that India can ill afford to turn its face away from.

**Q: How long does it take for technology to sink through?**
In my earlier years I was of the view that technology transfer takes time. I've been very pleasantly surprised for two reasons: the alacrity with which people in the railways designed the Vande Bharat train. I've travelled on the train myself from Delhi to Kanpur, and I can say it is 70 percent of what I would have expected from a Swiss or German train. It's completely indigenously built at one-eighth the cost at which foreign companies would have done it. And it didn't take long for our railway engineers, whom we criticize as being stodgy, to design and build it. The big picture is that the political class had confidence in them, empowered them, and we now have Vande Bharat.

My second point is there have been some issues with the Japanese because of the delay in the bullet train project. According to the grapevine they had kind of offered the next version, which was far more expensive than what we had budgeted for. The government has actually placed the order for two bullet trains, with a joint venture of Bharat Earth Movers Ltd (BEML) and the Integral Coach Factory. A ₹827-crore order has been given to this entity, two PSUs in partnership, to develop two bullet train rigs.

The fact is that we are almost there. Once the Japanese bullet trains come in, our guys are going to strip it down like the Chinese did decades ago. We have the scientists, we have the railway engineers, so I am far more sanguine about our ability to absorb technology in half a decade, be self-sufficient and start exporting at a reasonable cost to the developing world, to the African continent, and parts of Central Asia.

**Q: How much are you being heard? What is the resonance with the government?**
You know, to be honest, I am slightly perplexed. I am not getting a resounding yes. But it appears that the railway board or the railway establishment or even the political establishment has not yet announced publicly any intention of moving ahead on four or seven corridors of high-speed rail. I am not clear what is holding them back.

There comes a time in economic history when a movement up the technology ladder becomes inevitable. Is it that the railway board mandarins are very comfortably set in their existing bubble of doing things incrementally? You know, thinking we will go a little faster, we will do Vande Bharat, we will do station development, decongest, connect Kashmir, connect the Northeast.

My hypothesis is that the political class has influenced the thinking of the railway board in two ways. One is regional connectivity. You would have seen a lot of capex going into connecting Kashmir, the Northeast and border areas, which is a politically correct decision.

The second is station redevelopment. A large portion of the travelling public is just fed up with the dirty quality of our stations and is therefore moving to low-cost air services or Volvo coaches, etc. So station redevelopment seems a big item on the political agenda. It is also touch and feel development, so it gets votes.

The third, I think, is let's spread the Vande Bharat butter across the Indian toast before we embark on high-speed rail. I'm not saying this thinking is illogical.

Look at the speed of trains. The existing Rajdhanis do about 120 kilometres per hour. The Vande Bharat is designed to do 160 kmph, but at peak it touches 128 or130. It doesn't go faster because of the quality of tracks and because of lack of access control over large sections. You have cows coming in, you have people walking across.

The next is high-speed rail. If Vande Bharat is designed for max speeds of 160 or 170 kmph, high-speed rail is 200 to 240 kmph. Our Ahmedabad-Mumbai bullet train is 320 to 330 kmph. Above that is the highly specialized Maglev, or the magnetic levitating trains of China, which move at 350 kmph. So you have layers of segmentation of technology relevant to customer needs and the state of the economy.

The problem with Vande Bharat is that it is using the existing tracks. So we have the same old problem: while Vande Bharat wants to increase its speed, it is trailing a freight train or the Toofan Mail or the Howrah-Krishnanagar-Lalgola passenger trains. Their existing routes are crowded.

> **'Seven high-speed rail corridors of around 700 km each have the capacity to galvanize ₹20 lakh crore of investments and create a huge number of jobs India can ill-afford to turn away from.'**

Whereas high-speed rail requires a different track altogether so you're not constrained. My biggest criticism is if you are deprioritizing high-speed rail for political reasons, to further extend the Vande Bharat network, it's a good idea, but it doesn't do away with the constraint of existing tracks. What will we achieve? We will just lose another decade.

**Q: Okay, so what you need is a special effort with high-speed trains, which also fits this growing pattern of demand.**
Correct. And therefore, you know, like in air travel, you have the Udan Scheme where many sectors are subsidized by the government, or you get a viability gap fund. So, you are trying to get air travel and connectivity to integrate India's society so that more Hawaii *chappals* can travel — a fine idea.

The railways are not catering only to the construction worker going back home. Obviously, it's important and we should strive to improve the quality of services we're giving them. But there are layers of customers. While we are servicing one layer with traditional trains, there is the next layer, the Rajdhanis, Shatabdis and Dorontos. The third layer is Vande Bharat and at the apex is the high-speed rail. There are clearly defined customer segments. And the customer for high-speed rail is the one that you're losing the fastest. They're going by air, they're buying fancy cars, jetting along.

You see the number of people who go on driving holidays now. There are lovely cars and highways and comfortable Volvo inter-city buses. You don't have to be on the waiting list of the railways. You are losing your highest paying customers. You can give better services and reduce the

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₹64,000 crore subsidy you are giving which is actually benefiting your highest paying customers.

**Q: The railways are crucial to the development needs of the country. But you need a more market-oriented railways.**

Why did we set up NHAI (National Highways Authority of India)? After all, the Public Works Department (PWD) has existed for centuries. So, if you take that analogy, we needed a separate agency with separate public funding and PPP (public-private partnership) structuring to think afresh about the design of roads, alignments and not leave it to the PWD at the Central or state level. We saw India become a success story in roads and highways.

I'm arguing that we already have a High-Speed Rail Corporation (HSRC) which is implementing the bullet train project. Expand that corporation slowly by giving them more high-speed rail corridors. If you expect the existing railway board to suddenly wake up and declare that they are going to do high-speed rail, it will never happen.

You have an opportunity here for institutional reform where all corridors are given to high-speed rail. And we are gradually able to reduce the dominance of the railway system by the traditionalist mindset of the railway board.

The railway board has a tremendous amount of work to do. Northeast connectivity, moving cargo and moving passengers. So, allow a new system with a new mindset to come up.

My next point is public finance. High-speed rail has tremendous spin-offs for the geographies it passes through. A study by the London School of Economics, the London Business School and Hamburg University has pointed out that cities, across countries, that have a station on the map of a high-speed rail link have a city GDP growth 2.7 percent higher than a city that does not.

We are also looking at urban dispersion. Everybody can't live in Delhi and Mumbai, but now people can live in Surat, Baroda, Navsari. From 12 intermediate stations you can reach Mumbai and Ahmedabad in a jiffy.

The Centre is funding the entire Bullet Train Project. But for the Metro Rail projects, the Central government funds 50 percent, and the state the remaining 50 percent. For high-speed rail why does only the Centre have to fund it? We must have a system whereby rich cities, like Pune, Mumbai and Indore, with large cash reserves from their efficiencies and the state government must come forward and contribute for a particular segment of high-speed rail. It should not be only a Central subject.

**Q: The elephant in the room is the slow and haphazard process of urbanization. Much of what you're saying actually hinges on a national urbanization effort.**

I don't see it as having reached the collective consciousness of our policymakers: the linkages of high-speed rail to economic growth to urban dispersion away from the big metros to increasing livelihoods. All these things are really one part. You can't just look at high-speed rail only by seeing a train or speed. It's not that. It's a transformation of society.

I'm slightly worried that the totality of the economic benefit, the national benefit and societal transformation, urbanization, all these pieces are not being easily grasped by public policy mandarins. Why isn't NITI Aayog publishing paper after paper saying let's do this? Why isn't the Prime Minister's Economic Advisory Council saying it should be the next big idea?

**Q: If you were to set up three corridors, which cities would you connect?**

First, since Ahmedabad and Mumbai are already to be connected by bullet train, it's only logical to extend it to Delhi. It's the highest economic corridor in the country. So, my first corridor is Delhi-Rewari-Jaipur-Ajmer-Jodhpur-Ahmedabad-Mumbai, of which Ahmedabad-Mumbai is already connected.

> **'I am arguing that we expand the corporation for the bullet train by giving them more high-speed rail corridors.'**

Corridor number two: Chennai to Mumbai. It goes through a very high economic zone — Tirupati-Bengaluru-Tumkur-Davangere-Dharwad-Belagavi-Kolhapur-Satara-Pune-Navi Mumbai with a spur to Goa.

Third corridor: Delhi to Amritsar — Delhi-Sonepat-Panipat-Karnal-Ambala-Chandigarh- Ludhiana-Jalandhar-Amritsar.

Fourth corridor: Delhi to Kolkata — Delhi-Agra-Lucknow-Varanasi-Patna-Kolkata.

These are my four prioritized corridors. We have researched all of them in depth. The economics completely stands behind suggesting these four out of the seven or 10 that have been popularly talked about in railway establishments. We should get on with it.

**Q: For a lot of people, actually, train travel is preferable to air travel. Air travel has become quite clumsy. Nowadays you end up spending a whole day getting to some city. What is the time you would be saving on a high-speed train?**

Earlier, I'm told, there were eight direct flights from Delhi to Jaipur. Today there are only two. Apparently, there are no passengers opting to fly to Jaipur from Delhi. They are either driving down or taking the Shatabdi/Vande Bharat or the Volvo coaches. That answers your question because the distance is 250 kilometres from Delhi to Jaipur.

So the answer is, wherever there is a difference of up to around 400 kilometres you will probably see traffic shifting to high-speed rail. These would be corridors like Mumbai-Pune, Bengaluru-Chennai, Delhi-Jaipur. Station development is important. You've got to make entry-exit from stations easier, and give people a sense of being in an airport, or at least in a mall. Our station development in the metros is disastrous, but that needs improving.

**Q: High-speed rail requires a humongous amount of money altogether, especially to do it fast. What would we do here?**

You know, money actually is not the problem. The Mumbai-Ahmedabad bullet train corridor costs roughly ₹250 crore per kilometre. Whereas the Metro rail today costs ₹500 crore per kilometre. Yet, the Metros are able to tie up the funding. Nobody has said that they don't have financial closure. And that is only a city-level development.

The funding today is for long gestation infrastructure projects. The world of finance has matured. NHAI has shown us this. Not only have they raised funds, they have been monetizing with InvITs (Infrastructure Investment Trusts) and others and rolling the money back. India has created a developmental financial institution called NaBFID (National Bank for Financing Infrastructure and Development). We have pension funds, insurance companies. We hardly use them for infrastructure development. And this is nation building.

**Q: What about land acquisition? Is this also going to haunt plans for high-speed rail?**

It only haunts projects if you have a massive political opponent in a particular state. The land for the bullet train in Gujarat was acquired without any hassle. It was the opposition in Maharashtra that stalled it in that state. Today, the only reason land acquisition is a problem is when you have very strident, top-level political opposition.

One of the biggest expressways is the Mumbai-Nagpur corridor. Has anybody talked about a problem in land acquisition? In UP six expressways have been built by the Yogi Adityanath government. Has anybody said anything about land acquisition? There is the new Jewar airport, the Navi Mumbai airport, the Tata semiconductor factory in Dholera. At prices being offered today, which are four times the market value, a lot of people in agriculture are only too happy to put the land to some other use because it's a subsistence existence. Believe me, land acquisition is not a problem. It has been made into a problem for reasons of politics. ■